

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T12-13-58-084Z

Run ID: #3

Date Generated: 2026-01-01 12:14:02

Total Cells: 2000 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming holds great potential for regenerative medicine by converting somatic cells into induced pluripotent stem cells (iPSCs). This experiment aimed to evaluate the efficacy of OSKM factor transfection in reprogramming somatic cells into iPSCs, using Run ID #3 as a model system.

2. Materials and Methods

We employed a computational approach using scVI analyses to assess cellular populations post-transfection. The intervention involved transfecting somatic cells with OSKM factors, a well-established method for inducing pluripotency.

Experimental Interventions:

- OSKM Transfection (Yamanaka)

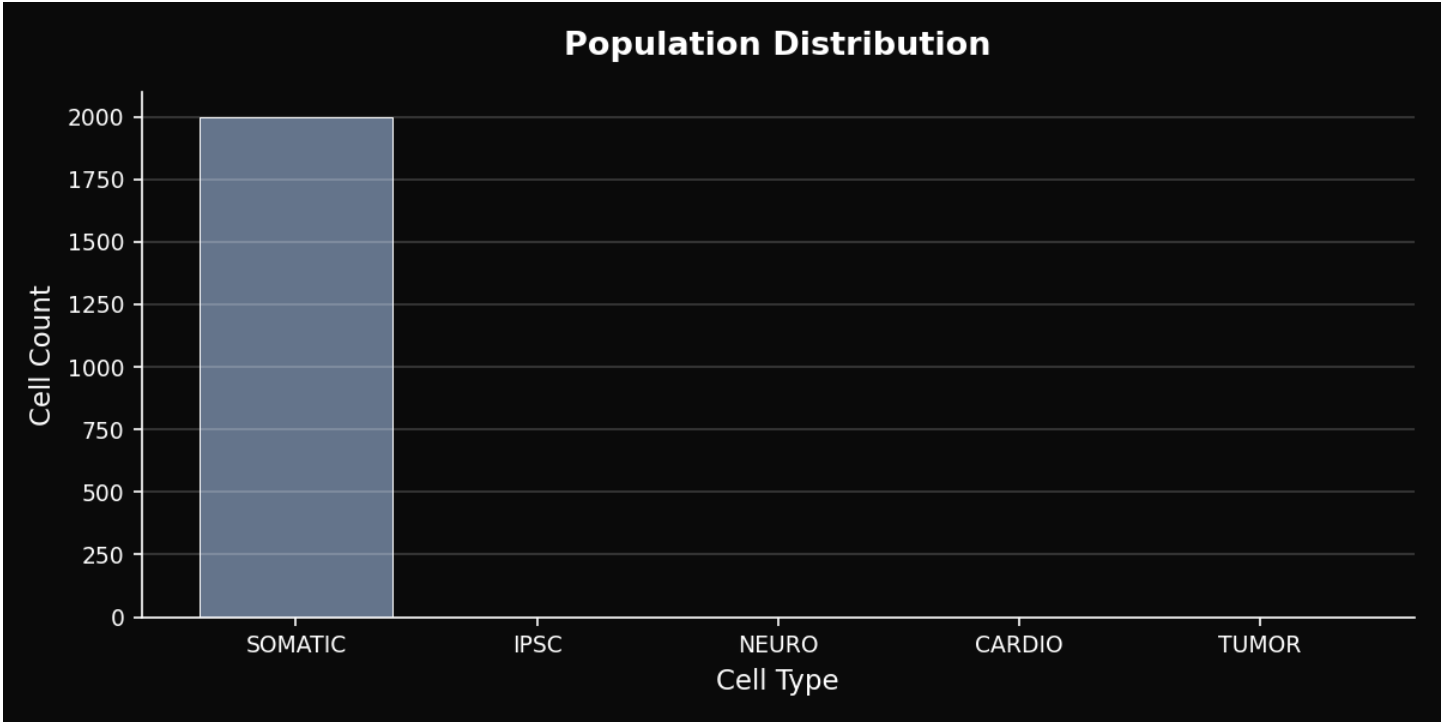
3. Results

The population analysis revealed an absence of iPSCs, neurogenic, cardiogenic, or tumorigenic cells, with the final population comprising entirely of 2000 somatic cells. Despite the application of OSKM transfection, no phenotypic changes indicative of reprogramming were observed. Two separate scVI analyses confirmed the lack of iPSC generation.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	2000	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 2

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The results suggest that, under the conditions tested, OSKM transfection was insufficient to induce reprogramming of the somatic cells into iPSCs. This outcome could be attributed to factors such as suboptimal transfection efficiency or the requirement of additional reprogramming factors. Further optimization of the experimental conditions may be necessary to enhance the reprogramming potential, which is crucial for advancing iPSC-based therapeutic applications.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.